

Data-Quality Findings Report

OpenAQ New Delhi Measurements — Pipeline & Quality Assessment

Extract: openaq_new_delhi_measurements.xls | Rows analyzed: 18,000

Prepared using the dataset obtained using OpenAQ Public API for location _ New Delhi

Executive Summary

- The extract silently blends two different monitoring deployments — sensors reporting from Feb 2016 and sensors reporting from Feb–Mar 2025 — with no field in the data to tell them apart. This is the single largest driver of the issues below.
- The NOx (nitrogen oxides) sensor stream is very likely mis-scaled: 99.7% of matched readings show NOx lower than NO alone, which is chemically impossible ($\text{NOx} = \text{NO} + \text{NO}_2$).
- Real data completeness is materially worse than the source's own metadata claims: coverage fields report 100% for all 18,000 rows, while independent reconstruction of the expected 15-minute cadence shows 42.1% of expected reading slots are missing overall.
- Three parameters (co, no2, so2) carry two different units across the merged extract, at risk of being silently averaged together incorrectly.
- Recommended immediate action: add a station identifier to the export, quarantine the NOx stream pending calibration review, and stop trusting the coverage metadata field as a completeness signal.

Dataset Scope

Period analyzed: Two disjoint windows: Feb 5–24, 2016 and Feb 18–Mar 6, 2025

Rows: 18,000 | **Sensors:** 18 (1,000 rows each) | **Parameters:** 12

Grain: One row = one sensor's reading for one 15-minute interval

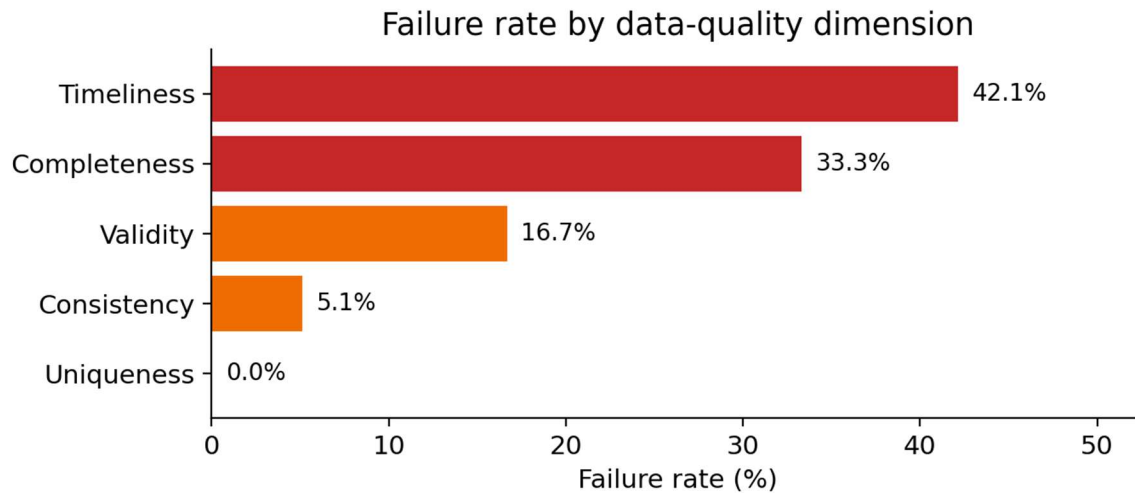
Source: Single flat-file OpenAQ export; no explicit station/location identifier field

Limitations: No external ground truth available to confirm the two-deployment hypothesis; see full assumptions log

Quality Results

Dimension	Evaluated	Failed	Failure rate	Severity
Completeness	54,000	18,000	33.3%	Critical/Medium (mixed)
Uniqueness	18,000	0	0.0%	— (pass)
Validity	36,000	6,000	16.7%	High
Consistency	18,972	969	5.1%	Critical
Timeliness	24,873	10,478	42.1%	High

Completeness blends two very different rules: value-not-null and key-fields-not-null pass at 100% — every reading that exists is usable. The always-null coordinates/summary fields are the sole driver of the 33.3% blended rate and are scored medium severity pending confirmation (Finding 4).



Major Findings

Finding 1 — Two monitoring deployments merged without a distinguishing field

Finding: Sensor IDs 35, 36, 392, 393, 394, 399 report data exclusively from Feb 5–24, 2016. Sensor IDs 12234782–12234790 and 14340713–14340715 report exclusively from Feb 18–Mar 6, 2025. No `station_id/location` field exists to confirm whether these are the same station or two different stations.

Evidence: Sensor-level profiling shows a complete, non-overlapping split by both sensor-ID range and date range.

Trend: Not applicable — a structural split, not a trend.

Affected population: All 18,000 rows carry this ambiguity; it directly drives Findings 2, 3 and 5.

Likely cause: Most consistent with a station relocation, hardware refresh, or an OpenAQ `location_id` change between 2016 and 2025 that was not carried into this export.

Business impact:

Business impact: Any trend or aggregate analysis treating this extract as one continuous series will silently compare two different deployments — 2016 pollutant levels (e.g. PM_{2.5} up to 637 µg/m³) are far higher than 2025 levels (up to 204 µg/m³), which could be misread as improvement when it may equally reflect different instrumentation, siting, or calibration.

Finding 2 — NO_x stream is very likely mis-scaled

Finding: By definition NO_x = NO + NO₂, so NO_x must always be ≥ NO alone. In this extract, 969 of 972 matched timestamps (99.7%) show NO_x below NO alone — roughly three orders of magnitude on average (mean NO_x = 0.040 ppb vs. mean NO+NO₂ = 62.5 ppb).

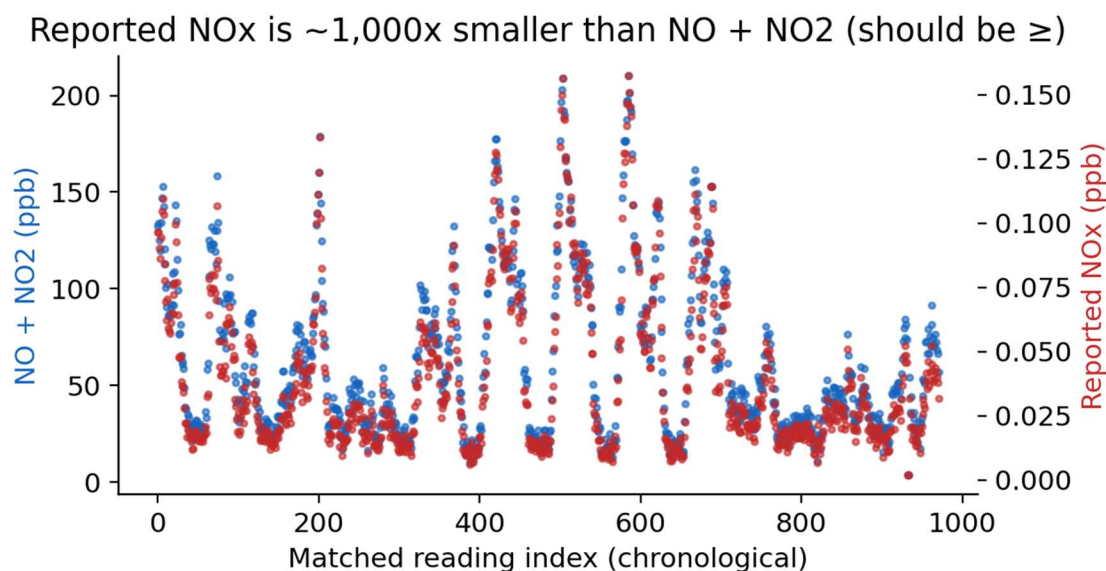
Evidence: Consistency check joining `no/no2/nox` on matching timestamps (2025 station only — 2016 station has no NO_x sensor).

Trend: Consistent across the full observed window for sensor 14340713; not a one-off spike.

Affected population: All 972 matched NO_x readings from sensor 14340713.

Likely cause: The magnitude and instability of the gap (expected-to-observed ratio ranges ~1,268×–1,900×, not a clean fixed factor) points to a sensor calibration or unit-labeling defect in the NO_x channel specifically. Reported as a likely contributing factor, not a confirmed root cause.

Business impact: Any report using this NO_x stream would materially understate nitrogen oxide levels.



Finding 3 — Source's own completeness metadata cannot be trusted

Finding: Every row carries coverage.percentComplete = 100.0 and percentCoverage = 100.0 — a constant with zero variation. Independently reconstructing the expected 15-minute cadence shows 10,478 of 24,873 expected slots (42.1%) are actually missing.

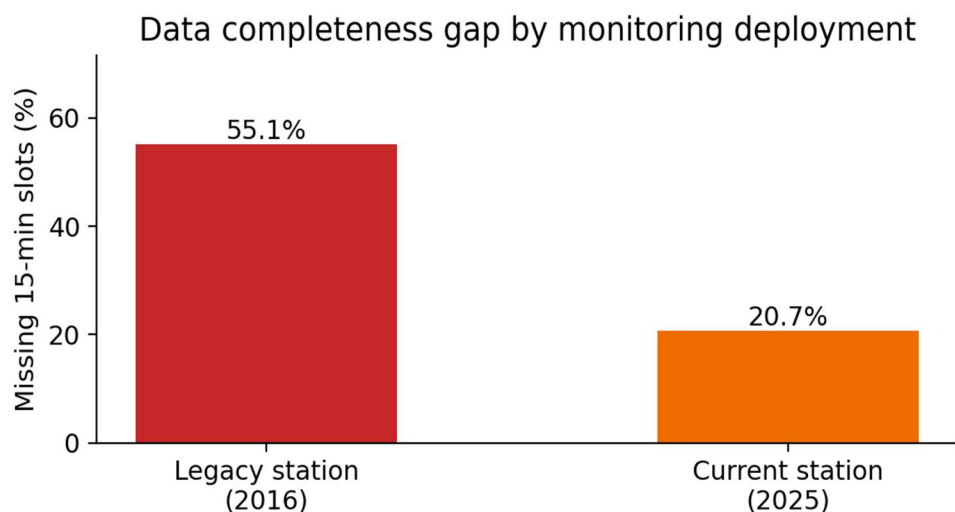
Evidence: Timeliness checks (DQ009 gap reconstruction, DQ010 metadata diagnostic).

Trend: Gaps are present throughout both monitoring windows, not concentrated at start/end; the legacy 2016 station is markedly worse (55.1% missing vs. 20.7% for the 2025 station).

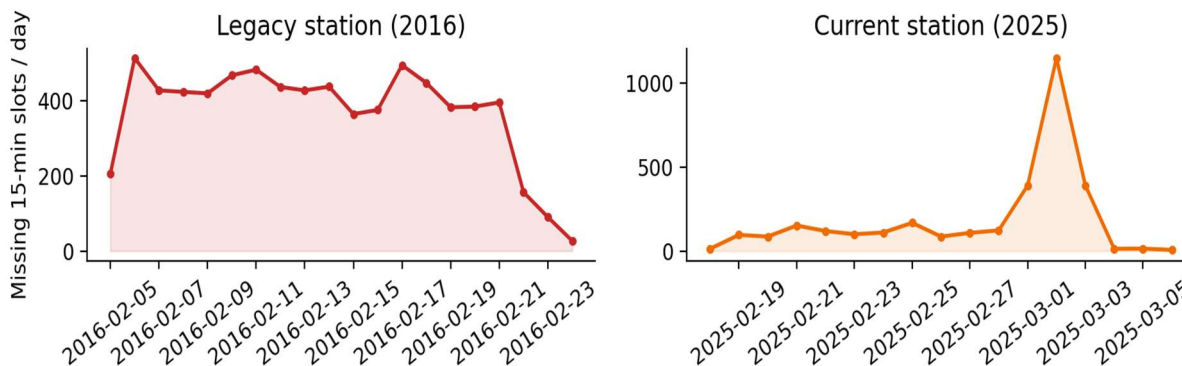
Affected population: All 18 sensors; more severe for the 6 legacy-station sensors.

Likely cause: The coverage block appears to describe completeness within the single 15-minute interval each row represents (1 of 1 expected sub-readings), not completeness of the overall time series — technically accurate for its own narrow definition but misleading as a series-level signal.

Business impact: A consumer trusting the coverage field at face value would believe this dataset has zero gaps, when more than two in five expected readings are absent.



Daily missing-slot volume — gaps persist throughout both windows, not just at the edges



Finding 4 — Structural completeness gap: coordinates and summary always null

Finding: coordinates and summary are NULL for all 18,000 rows — a complete absence, not a partial gap.

Evidence: Null profiling, confirmed against raw file cells (not a parsing artifact).

Trend: Constant throughout the extract.

Affected population: 100% of rows.

Root cause: Undetermined from this extract — could be an export-configuration choice or a genuine upstream gap.

Business impact: Without coordinates, this dataset cannot be mapped or geographically validated without an external join to station metadata.

Finding 5 — Three parameters carry two different units

Finding: co, no2, and so2 each appear under both ppb (2025 station) and $\mu\text{g}/\text{m}^3$ (2016 station) — 6,000 of 18,000 rows (33.3%) belong to a parameter with more than one unit in this extract.

Evidence: Unit-consistency validity check (DQ006).

Trend: Direct consequence of Finding 1 — present throughout both windows.

Affected population: 6,000 rows across co, no2 and so2.

Likely cause: Direct consequence of Finding 1 — the two deployments report in different units for the same parameters and nothing in the schema flags this.

Business impact: Averaging or summing these parameters across the full extract without unit conversion first would produce a meaningless blended figure differing by 2–3 orders of magnitude.

Root-Cause Analysis

Confirmed by evidence in this extract

- Two disjoint sensor-ID/date-range populations exist in one file (Finding 1)
- NOx is chemically inconsistent with NO/NO2 in the same file (Finding 2)
- The coverage metadata field is constant despite real, independently measurable gaps (Finding 3)
- Three parameters carry two units within the same file (Finding 5)

Likely contributing factors (not fully provable from this extract alone)

- The two-deployment split most likely reflects a station relocation or location_id change between 2016 and 2025 that a downstream export process did not distinguish

- The NOx defect is most consistent with a sensor calibration or firmware/unit-labeling issue specific to that channel

Unknowns

- Whether coordinates/summary are expected to populate for this export type at all
- The exact pipeline stage responsible for the NOx scaling defect

Recommendations

#	Recommendation	Owner	Priority
1	Add a station_id/location field to the export; confirm with the source owner whether the 2016 and 2025 sensor sets are the same physical station	Data engineering / pipeline owner	Critical
2	Quarantine the NOx stream (sensor 14340713) from reporting until calibration/unit labeling is reviewed	Sensor/instrumentation owner	Critical
3	Stop trusting coverage.percentComplete/percentCoverage as a completeness signal; replace with an independent 15-min cadence check and SLA alert	Data/analytics team	High
4	Investigate why the legacy 2016 station's gap rate (55.1%) is far worse than the current station (20.7%) before using it as a completeness benchmark	Data engineering	High
5	Confirm whether coordinates/summary are expected to populate for this endpoint; fix the export or remove the columns	Data engineering / source owner	Medium
6	Add a unit-normalization step (ppb ↔ µg/m³) for co, no2, so2 before any cross-deployment aggregation	Data/analytics team	High

Limitations

- No independent ground truth (e.g. an official OpenAQ API pull for the same window) was available to confirm whether the 2016/2025 split is two stations, a relocation, or a backfill artifact — flagged as a likely contributing factor, not a confirmed root cause.
- The NOx unit/scale defect could not be tied to a specific pipeline stage (sensor firmware, ingestion transform, or export step) from this extract alone.
- coordinates and summary being 100% null could reflect either a genuine upstream defect or a field this endpoint is not expected to populate — not resolvable from the data alone.
- Statistical anomaly thresholds use this extract as their own baseline (no external historical series was available), so legitimate seasonal pollution spikes may be indistinguishable from true anomalies without a longer baseline.
- No production incident response, pipeline rebuild, or ML-based detection was undertaken — this is an analytical diagnosis and recommendation only, per project scope.